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TECHNICAL PAPER

Particle Swarm Optimization–Based Machine Learning Algorithms for Developing the Modified Proctor Compaction Parameter Prediction Software

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Abstract

Maximum dry density (MDD) and optimum moisture content (OMC) are two significant compaction criteria, especially for quality control and design engineers. Estimating laboratory proctor compaction test is rigorous, time-consuming, and expensive, hindering projects with limited budgets and tight schedules. This study proposed the novel application of hybrid particle swarm optimization (PSO) optimized Gaussian process regression (GPR), K-nearest neighbor (KNN), random forest (RF), and extreme gradient boosting (XGB) algorithms for predicting the soil compaction parameters. Analyzing 2148 in situ soil samples from various geological locations established the maximum proficiency of the XGB algorithm followed by KNN, GPR, and RF in MDD, whereas XGB, KNN, RF, and GPR in OMC. Furthermore, the level 1 and level 2 validation results ascertain the robustness of models in predicting MDD and OMC on different geological location datasets. Eventually, the AI-based computer software developed through this study offers reliable and efficient predictions for civil engineers.

Keywords Machine learning algorithms · Hyperparameter optimization · Computer software · Modified compaction parameters · Borrow materials

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1 Introduction

1.1 Problem statement

Borrowed materials play a crucial role in various civil engineering projects like building foundations, trench backfilling, retaining walls, earth dams, railways, airfields, and highway subgrades (Khuntia et al., 2015; Farooq et al., 2016; Karimpour-Fard et al., 2019). The quantity and quality of these materials required for construction are determined using compaction parameters: maximum dry density (MDD) and optimum moisture content (OMC). Compaction serves three primary objectives: reducing permeability, minimizing settlement, and enhancing shear strength (Gurtug & Sridharan, 2002; Gurtug et al., 2018). In general, laboratory compaction parameters are achieved through the Proctor compaction test, which was developed by Proctor (Proctor, 1933). However, this method demands around 20 kg of soil (to obtain a minimum of six points on the compaction curve) for each trial and takes 2–3 days to complete. Preserving such a considerable quantity of soil in the laboratory and conducting several compaction tests becomes more expansive, rigorous, and time-consuming (Gurtug & Sridharan, 2004; Di Matteo et al., 2009; Patra et al., 2010a; Patra et al., 2010b; Bera & Ghosh, 2011). This challenge becomes more acute for field engineers, especially when locally available materials' properties vary frequently along small land stretches, necessitating numerous lab tests for material selection. As a result, project timelines are extended, leading to escalate the project costs (Viji et al., 2013; Verma & Kumar, 2022).

1.2 Literature Review

In the past years, several attempts have been made to predict the compaction parameters of various types of soils at different compactive energy levels through numerous computational approaches. To the author's knowledge, the first model in predicting the compaction parameters was established by (Ring III et al., 1962). They developed the correlation equations for MDD and OMC through soil classification data containing liquid limit (LL), plastic limit (PL), plasticity index (PI), fineness average (FA), approximate average particle diameter (D_{50}), and the content of particle finer than 0.001 mm as input parameters. Ramiah et al. (Ramiah et al., 1970) developed the correlation equation for clay soil through LL alone. Korfiatis and Manikopoulos (Korfiatis & Manikopoulos, 1982) developed the correlation for MDD of granular soils using specific gravity and fine content of soils. Blotz et al. (Blotz et al., 1998) presented the correlation equation for MDD of clay soil through LL. They also introduced various compaction energy levels such as reduced Proctor, standard Proctor, modified Proctor, and super-modified Proctor. Omar et al. (Omar et al., 2003) attempted to establish the predictive equation for MDD and OMC of 311 granular soil samples occupied from various locations in the United Arab Emirates. Di Matteo et al. (Di Matteo et al., 2009) tried to establish regression equations for estimating the modified compaction parameters of clay and fine-grained soil. From 126 soil sample datasets, Günaydın (Günaydın, 2009) observed that artificial neural network (ANN) approach is more superior to regression analysis. Das et al. (Das et al., 2011) adopted two AI approaches, i.e., functional networks (FNs) and multivariate adaptive regression splines (MARSs) in predicting the MDD and unconfined compressive strength (UCS) of cement-stabilized soil. Using ANN, least square support vector machine (LS-SVM), and MARS methods, Khuntia et al. (Khuntia et al., 2015) attempted to develop the correlation for MDD and OMC of 110 granular soil samples taken from Mujtaba et al. (Mujtaba et al., 2013) study. Nagaraj et al. (Nagaraj et al., 2015) introduced the concept of modified plastic limit in predicting the compaction parameters of soils. Based on the compaction energy, LL and PI parameters, Farooq et al. (Farooq et al., 2016) developed the correlation equation for quick estimation of compaction parameters. Suman et al. (Suman et al., 2016) used the various artificial neural network (ANN) models such as Bayesian regularization neural network (BRNN), Levenberg-Marquardt neural network (LMNN) and differential evolution neural network (DENN), and support vector machine (SVM). The proposed ANN equations could be used for predicting the MDD and UCS of cement-stabilized soil. Ardakani and Kordnaeij (Ardakani & Kordnaeij, 2017) observed that the group method of data handling (GMDH) is more reliable than the genetic algorithm (GA) in predicting the compaction parameters of soil. Using the results of 728 dataset (from his own study and literature dataset), Karimpour-Fard et al. (Karimpour-Fard et al., 2019) observed that ANN approach is considerably effective as compared to regression analysis in developing the prediction model for compaction parameters of soil. From a deep insight review of past literature studies, Verma and Kumar (Verma & Kumar, 2019) observed that compaction parameters of coarse-grained soil are significantly influenced by gradational parameters, whereas Atterberg's limits were found more essential for fine-grained soils. Using 451 soil datasets, Kurnaz and Kaya (Kurnaz & Kaya, 2020) establish that extreme learning machine (ELM) technique is considerably superior to BRNN, SVM, and GMDH.

1.3 Literature Gap

The literature review (also summarized in Table 1) demonstrates a balanced usage of regression techniques and artificial intelligence (AI) approaches by researchers in the past. Additionally, a substantial portion of researchers manually fine-tuned hyperparameters using time-consuming trial and error methods, which is particularly impractical for high-dimensional datasets. Frequently, data related to soil classification and compaction parameters display high imbalance and nonlinearity. As a result, conventional statistical methods and manual tuning often yield unsatisfactory outcomes. Several recent published articles (Omran et al., 2016; Kirts et al., 2018; Wang & Yin, 2020; Alzabeebee et al., 2021; Eleyedath et al., 2021; Jalal et al., 2021; Kardani et al., 2021a; Eleyedath & Swamy, 2022; Jacobsen & Teizer, 2022; Singh et al., 2022) underscore the reliability of AI-based prediction models in civil engineering.

Notably, prior studies predominantly centered on the standard Proctor test, which proves unsuitable for numerous heavy construction project works. Although, Omar et al. (Omar et al., 2003), Di Matteo et al. (Di Matteo et al., 2009), Khuntia et al.

Table 1 Description of literature studies performed to predict MDD and OMC of various soil types

S. No.	Literature study	Number of dataset	Type of soil	Compaction energy level	Computational approach	Hyperparameters tuning method
1	Al-Khafaji (Al-Khafaji, 1993)	88	Mixed	Standard	Curve fitting technique	NA
2	Omar et al. (Omar et al., 2003)	311	Granular	Modified	Regression	NA
3	Sridharan and Nagaraj (Sridharan & Nagaraj, 2005)	64	Fine-grained	Standard	Regression	NA
4	Di Matteo et al. (Di Matteo et al., 2009)	71	Clay, Fine-grained	Modified	Regression	NA
5	Günaydın (Günaydın, 2009)	126	Mixed	NA	Regression ANN	Manual tuning for ANN
6	Khuntia et al. (Khuntia et al., 2015)	110	Coarse-grained	Standard, Modified	MARS ANN LS-SVM	Manual tuning
7	Nagaraj et al. (Nagaraj et al., 2015)	57	Natural	Standard	Regression	NA
8	Farooq et al. (Farooq et al., 2016)	105	Fine-grained	Standard, Modified	Regression	NA
9	Ardakani and Kordnaeij (Ardakani & Kordnaeij, 2017)	212	Mixed	NA	GMDH GA	Manual tuning
10	Karimpour-Fard et al. (Karimpour-Fard et al., 2019)	728	Mixed	Standard	Regression ANN	Manual tuning for ANN
11	Kurnaz and Kaya (Kurnaz & Kaya, 2020)	451	Mixed	Standard	ELM SVM GMDH BRNN	Manual tuning
12	Present study	2148	Mixed	Modified	GPR KNN RF XGB	Particle swarm optimization

(Khuntia et al., 2015), and Farooq et al. (Farooq et al., 2016) attempted to formulate models for modified Proctor compaction parameters, their efforts were limited by parameter ranges, dataset sizes, statistical methodologies, and soil-type specificity. Additionally, these models were exclusively developed and validated for particular geological locations, making them inadequate for broader use. Consequently, relying on these models can lead to confusions and erroneous conclusions, which inhibit the generalization ability of the models. Recently, the validation results of Farooq et al. (Farooq et al., 2016) model shown by Verma and Kumar (Verma & Kumar, 2022) also states that in addition to LL, PL, PI, and grain size distribution parameters, MDD and OMC are an explicit function of geological location as well. Furthermore, Nagaraj and Suresh (Nagaraj & Suresh, 2018) and Verma et al. (Verma et al., 2023) demonstrated the substantial impact of soil mineralogical composition on geotechnical properties, which varies across locations. This emphasizes the need for a comprehensive AI-based prediction model tailored to compaction parameters for heavy construction project works. Such a model should encompass a wide range of properties, a substantial dataset, and strong generalizability from a geological perspective.

1.4 Research Significance and Contributions

This study aims to propose a software for predicting one of the foremost problem of quality control and design engineers, i.e., estimation of MDD and OMC of soils. For this purpose, numerous machine learning algorithms such as Gaussian process regression (GPR), K-nearest neighbor (KNN), random forest (RF), and extreme gradient boosting (XGB) were adopted. Notably, this study introduced a novel approach by utilizing particle swarm optimization (PSO), a swarm intelligence–based optimization technique, for hyperparameters tuning of these algorithms in predicting MDD and OMC of soil. The research utilizes a dataset of 2148 in situ soil samples encompassing a wide array of geotechnical properties, such as Atterberg's limits, grain size distribution parameters, and compaction parameters. This dataset was not only collected from the author's own study area but also sourced from diverse geological locations worldwide, as documented in past literature studies. The K-fold data division approach was adopted to strategically allocating the datasets for training purposes. Once the model was developed, its real-world applicability was recognized through level 1 and level 2 validating processes. Ultimately, laborless and financially feasible computer software for predicting MDD and OMC of soil was generated for assistance to the field engineers. The overall performance of developed models reveals that the study could be significant for the field engineers especially for the projects where there is a lack of equipment, limited time frame, and financial restrictions. However, further modification in the model or computer software could also be done by incorporating the dataset from other geological locations/country.

1.5 Applied Machine Learning Algorithms

1.5.1 Particle Swarm Optimization

Particle swarm optimization (PSO) is a metaheuristic optimization approach based on the simulation of the social behavior of birds within the flocks. It was proposed by Kennedy and Eberhart (Kennedy & Eberhart, 1995) which was later modified by Shi and Eberhart (Shi & Eberhart, 1998). This algorithm has numerous benefits, including (a) requires little memory for computing purposes; (b) the capacity of being understood and implemented in an easy and quick way; and (c) containing fewer parameters in need of adjustment compared to other evolutionary algorithms (Murlidhar et al., 2020). In PSO algorithm, to find the best structural position, every solution is presented as a particle that searches the best global (g_{best}) and the best personal (p_{best}) positions, based on its best solution (Dimou & Koumousis, 2009; Hasanipanah et al., 2017; Inti & Tandon, 2017; Moayedi et al., 2020; Roy & Singh, 2020). During the PSO modeling, each particle updates their velocity and position, using Eqs. (1) and (2), respectively, at each of the iteration. The velocity is updated based on the previous best position of the current particle (influence of particle memory) and previous global best solution found from the whole population (swarm influence).

Let, p_k^i and v_k^i indicate the position and velocity of particle k at iteration i, respectively. Then, the velocity and position of particle k at the next iteration is calculated by following formulas:

$$v_k^{(i+1)} = \omega v_k^i + c_1 r_1 \left(pbest_k - p_k^i\right) + c_2 r_2 \left(gbest_k - p_k^i\right) \tag{1}$$

$$p_k^{(i+1)} = p_k^i + v_k^{(i+1)} \tag{2}$$

where ω is the inertia weight of present particle used to modify the next generation of particle. r_1 and r_2 are uniformly distributed random numbers in the interval [0,1]. c_1 and c_2 are the acceleration coefficients used to handle exploitation and exploration capability. The group of particles/swarms cooperate with each other in obtaining an optimal solution with pre-defined best fitness function (this study adopted mean absolute error). The basic steps of PSO algorithm are shown below (Kennedy & Eberhart, 1995).

Step 1. Set iteration value $i = 0$ and generate the particle at random position p_k and velocity v_k vectors for initialization.

Step 2. Evaluate the fitness value for each generated particle.

Step 3. Check and update the values for $p$best and $g$best.

Step 4. For each particle, evaluate and update the velocity using Eq. (1).

Step 5. For each particle, evaluate and update the position using Eq. (2).

Step 6. Modify $i = i + 1$ and iterate again from step 2, until the termination criteria is satisfied.

1.5.2 K-Nearest Neighbor

K-nearest neighbor (K-NN) is a non-parametric supervised ML algorithm that uses the *k*-number of most similar outputs from the TR dataset (Peterson, 2009; Amiri et al., 2016; Chen & Hao, 2017; Hsieh, 2021). The KNN algorithm is also known as a lazy learner algorithm, because, instead of immediate prediction, it stores the dataset and categorize it based on their similarities then approaches an action on the dataset. KNN predicts the new records/data for the regression and classification-type problems based on their Euclidean distances, estimated mean, median, or model output variable (Yu et al., 2016; Cheng & Hoang, 2016; Oh et al., 2015; Kang et al., 2021).

1.5.3 Gaussian Process Regression

Gaussian process regression (GPR) is a probabilistic, nonparametric Bayesian approach for generalizing the nonlinear and complex problems related to regression and classification-type datasets (Wang, 2020; Dutta et al., 2018). Many supervised ML algorithms learn exact values from the dataset, whereas GPR infers a probability distribution of all admissible functions that could reasonably fit the data space regarding the problems (Ly et al., 2022). GPR is very efficient to handle nonlinear data due to the use of kernel functions. A GPR model can make predictions by incorporating the prior knowledge through covariance (kernel) functions and provide uncertainty measures over predictions (Dutta et al., 2018; Dao et al., 2020; Ghanizadeh et al., 2021; Williams & Rasmussen, 2006). The algorithm has recently received huge attention of researchers as of having the ability to solve many complex engineering problems related to various disciplines (Dutta et al., 2018; Ly et al., 2022; Ghanizadeh et al., 2021; Cai et al., 2020; Ceylan, 2020; Zeng et al., 2020; García-Nieto et al., 2021).

1.5.4 Random Forest

Random forest (RF), proposed by Breiman (Breiman, 2001), is a popular ML algorithm that belongs to a class of supervised learning technique. It is based on the concept of ensemble learning which means that instead of relying on a single decision tree, it takes the prediction from multiple learner and combines them to solve a complex problem (Zhou et al., 2019; Zhang et al., 2021). For instance, there is an input dataset $D = d_1, d_2, d_3, d_4, \ldots\ldots\ldots y_n$ where n denotes the number of predictive variables. The RF model would be a set of T trees $T_1(D)$, $T_2(D)$, $T_3(D)$, $T_4(D)$, $\ldots\ldots\ldots T_n(D)$. The prediction results of these decision making trees is Y_1, Y_2, Y_3, Y_4, $\ldots\ldots\ldots Y_n$. The final outcomes of RF model is the average of all the above decision trees (Yuchi et al., 2019; Karir et al., 2022).

1.5.5 Extreme Gradient Boosting

Extreme gradient boosting (XGB) is a decision tree–based ensemble ML algorithm developed by Chen et al. (Chen et al., 2015). Initially, XGB is based on gradient boosting architecture (Friedman, 2001), which uses various complement functions to estimate the results using Eq. (3).

$$\overline{y_i} = y_i^0 + n\sum_{j=1}^{n} f_j\left(S_i\right) \tag{3}$$

where $\overline{y_i}$ denotes the predicted output for the ith data with the parameter vector S_i; n denotes the number of estimators corresponding to independent tree structures for each f_j; y_i^0 is the primary hypothesis, i.e., mean of the original parameters in the TR dataset; η is the learning rate.

According to Eq. (3) in the jth stage, the jth estimator is connected to the model and the prediction of the jth. y_i^{-j} is calculated from the estimated output $y_i^{-(j-1)}$ in the next step, and the established f_j of the jth complementary estimator is shown in (4).

$$y_i^{-j} = y_i^{-(j-1)} + \eta f_j \tag{4}$$

where f_k represents the leaf weight that is established by reducing the objective function of the jth tree and is given by Eq. (5).

$$f_{obj.} = \gamma Z + \sum_{k=1}^{m}\left[g_k\omega_k + \frac{1}{2}\left(h_k + \lambda\right)\omega_k^2\right] \tag{5}$$

where Z denotes the leaf nodes quantity, λ denotes constant coefficient, γ indicates the complexity parameter, ω_k^2 indicates the leaf weight from 1 to Z, g_k and h_k are the summation parameters for the entire dataset associated with k leaf of the initial and previous loss function gradient, respectively.

In order to build the jth tree, a leaf is distributed into several leaves. Such a system is implied by using the gain parameters which is expressed through Eq. (6).

$$G = \frac{1}{2}\left[\frac{X_L^2}{Y_L + \lambda} + \frac{X_R^2}{Y_R + \lambda} + \frac{\left(X_L + X_R\right)^2}{Y_L + Y_R + \lambda}\right] \tag{6}$$

1.6 Data Description and Preparation

1.6.1 Description of Collected Dataset

Over almost 4 years, a comprehensive database of soil with modified proctor compaction properties was collected from different locations in India. Apart from the author's own study area, the rigorous literature review dataset was also collected worldwide. Table 2 tabulates the geological locations along with the amount of dataset used from these literature studies. Among the 1782 dataset of the author's study, 1086 were taken from the Varanasi Gorakhpur section of NH-29, 299 from Varanasi Babatpur airport ring road, 268 from Purvanchal Expressway which belongs to the state of Uttar Pradesh (UP), and the rest 129 were from Delhi-Gurugram, India. The

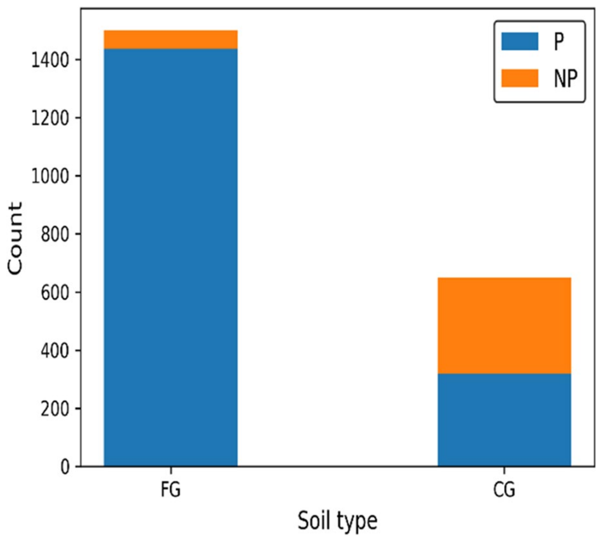
Table 2 Description of dataset collected for the analysis

S. no.	Author's study	No. of soil samples	Geological locations
1	Kin (Kin, 2006)	56	Malaysia
2	Yildirim and Gunaydin (Yildirim & Gunaydin, 2011)	124	Turkey
3	Yared (Yared, 2013)	42	Ethiopia
4	Farias et al. (Farias et al., 2018)	96	Peru
5	Katte et al. (Katte et al., 2019)	33	Cameroon
6	Gül and Çayir (Gül & Çayir, 2020)	15	Turkey
7	Author's own study	1782	India
Total dataset		2148	

complete soil dataset is comprising of fine-grained (FG) and coarse-grained (CG) soil of plastic (P) and non-plastic (NP) in nature as shown in Fig. 1.

The collected dataset contains information about the liquid limit (LL), plastic limit (PL) and plasticity index (PI), gravel content (G), sand content (S), fine content (FC), maximum dry density (MDD), and optimum moisture content (OMC) of soils (Fig. 2). Descriptive statistic values for the abovementioned parameters are tabulated in Table 3. This shows that the selected dataset covers an extensive range of index and engineering properties. However, for the detailed investigation, frequency distribution analysis was performed and the obtained result for the geotechnical parameters is shown in the form of histogram plot in Fig. 3. It is observed from Fig. 3 that the percentage of gravel varies from 0 to 94% and the maximum observations exists up to 10%, and sand content varies from 0 to 100% and was reported uniformly

Fig. 1 Type of soil used for developing the prediction model



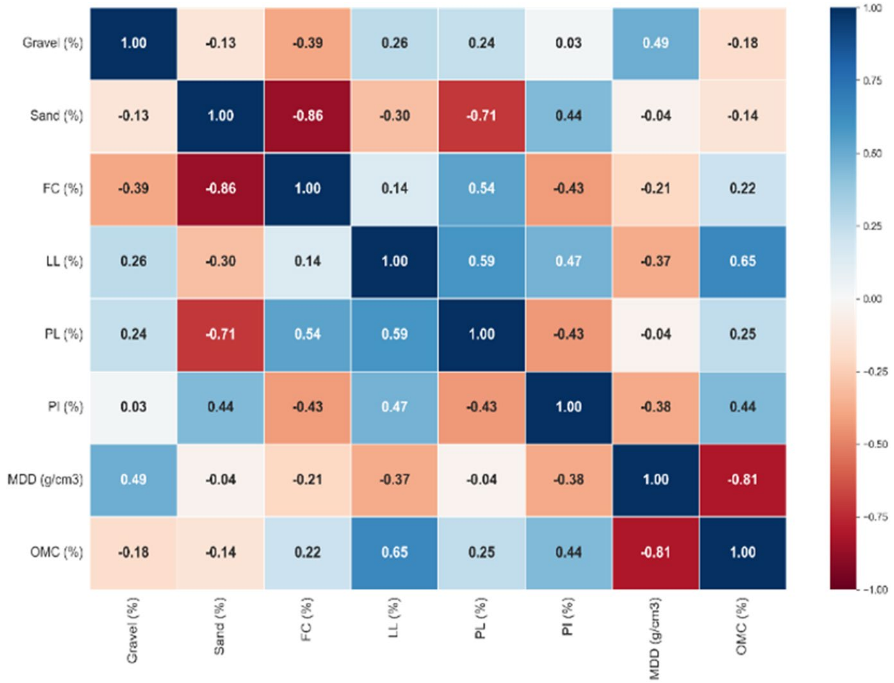


Fig. 2 Correlation matrix for all the geotechnical parameters

Table 3 Descriptive statistic values for all geotechnical parameters

	G (%)	S (%)	FC (%)	LL (%)	PL (%)	PI (%)	MDD (g/cm^3)	OMC (%)
Min	0.00	0.90	0.00	15.00	0.00	0.52	1.210	4.50
Max	94.00	100.00	99.10	98.00	62.10	72.00	2.330	40.20
Range	94.00	99.10	99.10	83.00	62.10	71.48	1.120	35.70
Average	5.67	28.11	66.22	30.45	18.01	12.45	1.855	12.42
Median	1.08	14.65	80.48	28.50	20.98	7.90	1.868	11.80
Mode	0.00	24.00	95.00	29.00	0.00	11.00	1.900	12.00
S.D.	13.42	24.46	26.30	9.69	9.50	8.69	0.115	3.44

throughout the ranges; almost similar findings were obtained for fine content which also varies from 0 to 100%. Liquid limit lies in the range of 15 to 100%, and values were frequently obtained from 20 to 30%; plastic limit and plasticity index were reported maximum between 0 to 30%. MDD and OMC were frequently observed between 1.5 and 2.0 g/cm^3 and 10 to 20%, respectively.

Pearson's correlation coefficient (R) measures the strength of the association between two parameters. The coefficient value varies from + 1 to − 1, where + 1 indicates the perfect positive correlation, zero denotes the no linear relationship, and

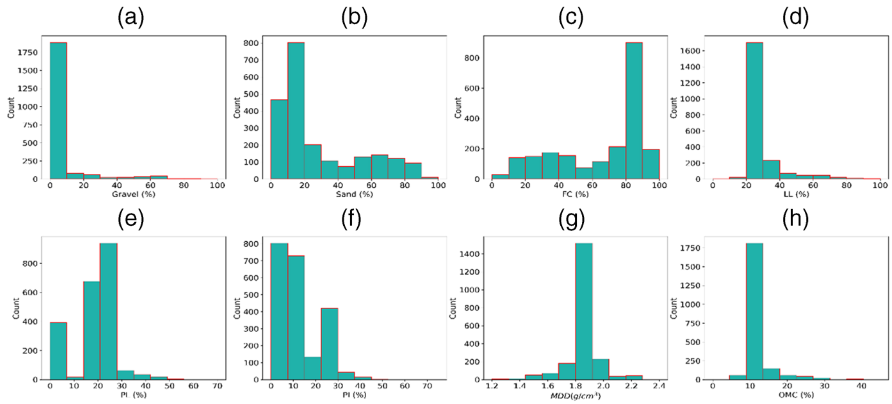


Fig. 3 Histogram plot for various geotechnical parameters

− 1 represents the perfect negative relationship. Furthermore, the type of relationship between the parameters along with the range of *R* value is presented in Table 4. Figure 2 shows the interrelationship obtained between all the geotechnical parameters in the form of the correlation matrix. As observed from Fig. 2, MDD and OMC are weakly and moderately associated with gravel, FC, and PL and LL and PI, respectively, whereas no correlation or very poor was found for sand content. The selected input parameters are gravel, FC, and PI. Further, the linear trend of MDD and OMC with these parameters is shown in Figs. 4 and 5, respectively.

1.6.2 Data Processing and Preparation for Analysis

In soft computing applications, normalization is recommended to enhance computational efficiency, such as decreasing the amount of time and memory required to solve a problem through particular algorithms. Normalization of the database is the process of bringing the required number of parameters within a predefined common scale without distorting the main features of the parameters. Using Eq. 7, all input and output parameters were normalized between 0 (lower limit) and 1 (upper limit), where *P* presents the parameters to be normalized; and $P_{\min}$ and $P_{\max}$ are the minimum and maximum values of the corresponding parameters, respectively.

$$P_{norm} = \frac{\left(P - P_{min}\right)}{\left(P_{max} - P_{min}\right)} \tag{7}$$

Table 4 Type of relationship between the parameters

R value	± 0 to ± 0.19	± 0.20 to ± 0.39	± 0.40 to ± 0.59	± 0.60 to ± 0.79	± 0.80 to ± 1.0
Relationship type	Very poor	Weak	Moderate	Strong	Very strong

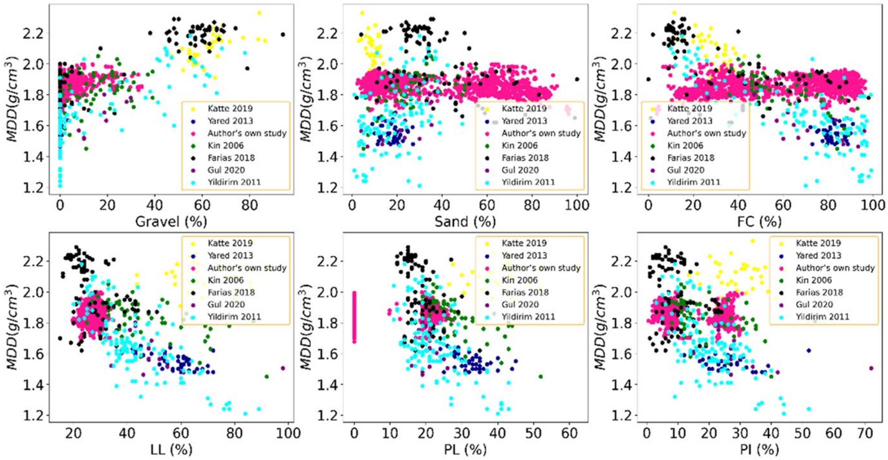


Fig. 4 Correlation plot of MDD with other geotechnical parameters

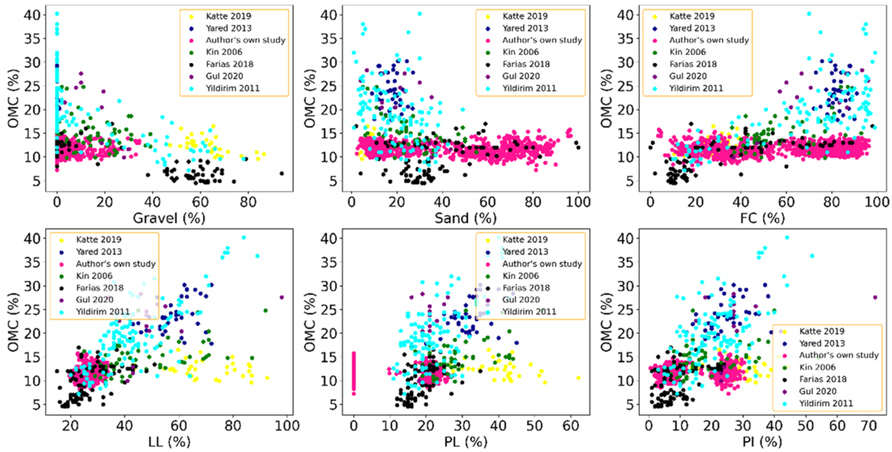


Fig. 5 Correlation plot of OMC with other geotechnical parameters

In the field of ML, it has been observed often that an improperly proportioned dataset may be responsible for vanishing some of the significant features of the training dataset, which may lead to under-fitting and over-fitting of the model and subsequently the accuracy of any predictive models. The *K*-fold data division approach was adopted to identify that which observations to be considered for training the model. Initially, the complete dataset was divided into training (TR) and testing (TS) sets. Among a total of 2148 observations, 75% (1611 datasets) were considered for training purposes and the rest 25% (537 datasets) were kept for testing the model.

This study used seven statistical performance measurement parameters such as coefficient of determination (R^2), coefficient of correlation (R), mean absoluter error

(MAE), root mean square error (RMSE), index of agreement (IOA), index of scatter (IOS), and a5 index for identifying the performance of the models. These parameters are formulated through Eqs. (8) to (14), respectively.

$$R^2 = 1 - \frac{\sum_{i=1}^{N}\left(y_i(a) - y_i(p)\right)^2}{\sum_{i=1}^{N}\left(y_i(a) - \overline{y_i(a)}\right)^2} \tag{8}$$

$$R = \frac{\sum_{i=1}^{N}\left(\left(y_i(a) - \overline{y_i(a)}\right)\left(y_i(p) - \overline{y_i(p)}\right)\right)}{\sqrt{\left(y_i(a) - \overline{y_i(a)}\right)^2\left(y_i(p) - \overline{y_i(p)}\right)^2}} \tag{9}$$

$$MAE = \left[\frac{1}{N}\sum_{i=1}^{N}\left|y_i(a) - y_i(p)\right|\right] \tag{10}$$

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i(a) - y_i(p)\right)^2} \tag{11}$$

$$IOA = 1 - \frac{\sum_{i=1}^{N}\left(y_i(a) - y_i(p)\right)^2}{\sum_{i=1}^{N}\left(\left|y_i(p) - \overline{y_i(a)}\right| + \left|y_i(a) - \overline{y_i(a)}\right|\right)^2} \tag{12}$$

$$IOS = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i(a) - y_i(p)\right)^2}}{\overline{y_i(p)}} \tag{13}$$

$$a10 - index = \frac{n10}{N} \times 100 \tag{14}$$

where $y_i(a)$ = actual value (laboratory obtained value); $y_i(p)$ = predicted value (value obtained through the developed model); $\overline{y_i(a)}$ = mean of actual value; $\overline{y_i(p)}$ = mean of predicted value; and N = number of observations.

1.7 ML Algorithm Hyperparameter Tuning

Using the TR dataset, various hyperparameters of GPR, KNN, RF, and XGB algorithms were optimized through the PSO algorithm. PSO requires a range of hyperparameters and then compares the fitness function values at each iteration. In this study, the number of iterations and the population of particles was set to 200 and 20, respectively. The termination criteria of the PSO algorithm was considered as number of iterations reached to the maximum number of iterations. The step-by-step methodology for developing the prediction model is shown in Fig. 6.

The most significant hyperparameters of GPR, KNN, RF, and XGB algorithms were tuned using PSO. The default values were taken for the other hyperparameters. In order to enhance the tuning efficiency, the search ranges of hyperparameters were specified based on few preliminary trials, like, some arbitrary unreasonable values can be bypassed. The search ranges along with the optimal values of these hyperparameters for MDD and OMC model are summarized in Table 5.

2 Results and discussion

This section sequentially illustrates the visual interpretation through scatter plot, violin plot, error frequency plot, regression error characteristic (REC) curve, and the overall performance of GPR, KNN, RF, and XGB models in terms of statistical measurement parameters and selection of best-fitted model through ranking analysis, respectively. Furthermore, two levels of validation were performed to identify the model robustness. Lastly, a simple and effective user-friendly prediction software was generated for the easiness to the field and laboratory engineers.

2.1 Visual Interpretation of Developed ML Models

Figures 7 and 8 depict the actual versus predicted value of GPR, KNN, RF, and XGB model in the TR and TS dataset of MDD and OMC, respectively. The center line represents the 1:1 line whereas the upper and lower line denotes the upper and lower bound which is taken as ± 5% and ± 20% for MDD and OMC, respectively. It is observed from Figs. 7 and 8 that the maximum number of data are distributed very close to the line of equality (1:1 line) or within the predefined upper and lower bound of MDD and OMC. Moreover, GPR, KNN, RF, and XGB models follow a specific and almost similar trend in MDD and OMC; however, the congestion of the dataset, within the bound and toward the line of equality, is maximum for the XGB

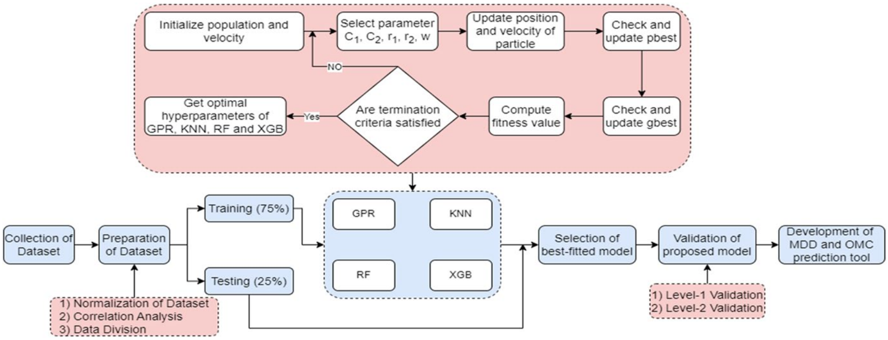


Fig. 6 Flow diagram for developing the hybrid PSO optimized ML models

Table 5 Optimal combination of ML algorithm's hyperparameters for MDD and OMC model

Hyperparameters	Tuning range	Optimal parameter setting (MDD)	Optimal parameter setting (OMC)
XGB			
Base score	0.1–0.9	0.612	0.162
Learning rate	0.01–0.3	0.048	0.1
Maximum depth	2–10	5	4
N estimators	10–100	59	23
Subsample	0.1–0.9	0.384	0.473
RF			
N estimators	10–100	24	47
Maximum depth	2–10	4	5
Minimum sample split	2–10	3	3
Minimum sample leaf	5–20	8	12
Maximum samples	0.1–0.5	0.245	0.347
GPR			
Kernel	RBF, rational quadratic and exponential sine squared	RBF (length scale = 1.5)	RBF (length scale = 5.2)
Alpha	1e−1 to 1e−20	1e−10	1e−10
KNN			
N neighbor	4–50	10	20
Leaf size	10–50	30	30
Distances	Euclidean, Manhattan, and Minkowski	Minkowski	Minkowski
Leaf size	10–100	30	30

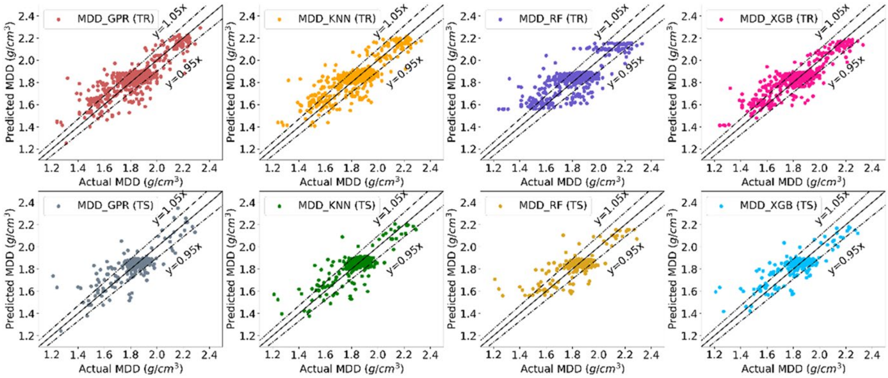


Fig. 7 Scatter plot for GPR, KNN, RF, and XGB model in TR and TS dataset of MDD

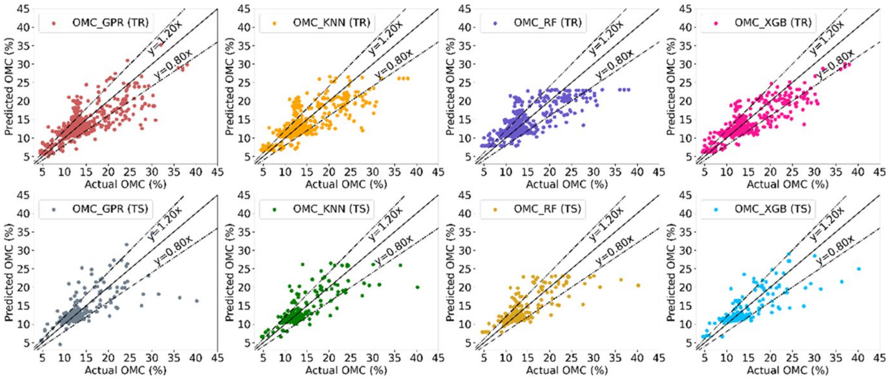


Fig. 8 Scatter plot for GPR, KNN, RF, and XGB model in TR and TS dataset of OMC

model. Therefore, the developed models for MDD and OMC are adequately fitted and generalization performances.

The full distribution of error obtained for GPR, KNN, RF, and XGB models in the TR and TS dataset of MDD and OMC are presented in the form of a violin plot which is similar to the box plot but with a combination of rotated density plot on each side. The left and right side portions of the violin plot are mutually mirror image. Figures 9 and 10 present the violin plot for the GPR, KNN, RF, and XGB models in the TR and TS dataset of MDD and OMC, respectively. The wider and thinner section of the plot signifies the higher probability density or frequency of the dataset, and the skinnier section represents a lower frequency. The top and lower nib of the plot indicate the range of the error value. As seen from Fig. 9, the minimum error is obtained for the XGB model in both TR and TS datasets, i.e., almost varies from + 15 and − 35%. Moreover, the wideness and thickness of the section in the mid of the plot demonstrate that MDD of soil can be predicted within ± 5% variations. This can also be confirmed through Fig. 11 showing an error bar plot for the GPR, KNN, RF, and XGB models in the TR and TS datasets of MDD. As seen from

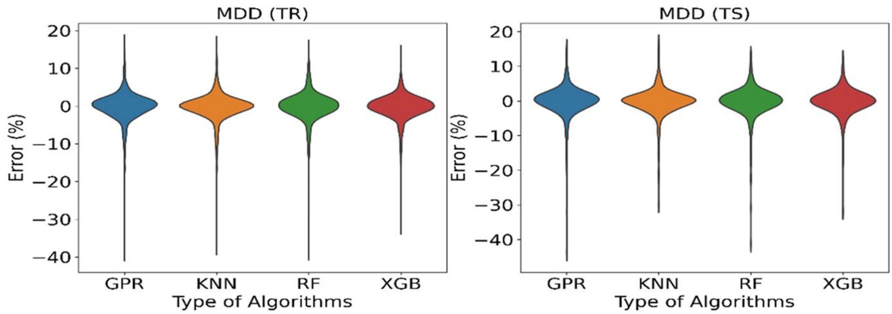


Fig. 9 Violin plot for GPR, KNN, RF, and XGB model in TR and TS dataset of MDD

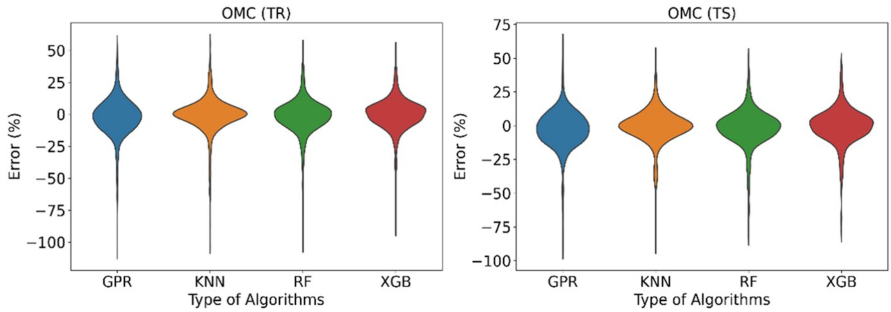


Fig. 10 Violin plot for GPR, KNN, RF, and XGB model in TR and TS dataset of OMC

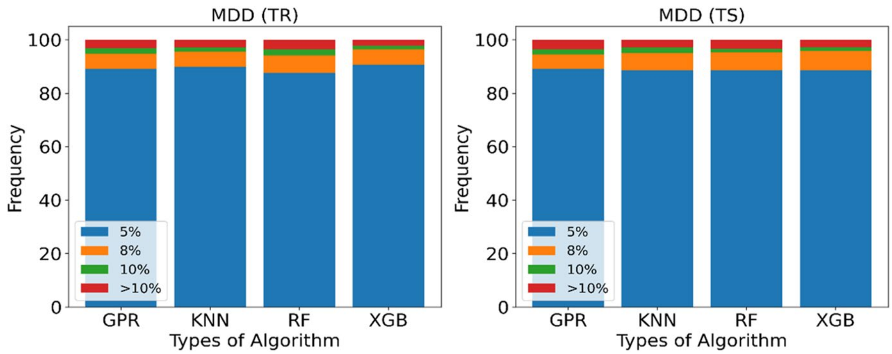


Fig. 11 Error frequency plot for GPR, KNN, RF, and XGB model in TR and TS dataset of MDD

Fig. 11, almost 90% observations of MDD can be predicted within ± 5% variations. Like MDD, the XGB algorithm is substantially adequate in predicting the OMC of soil as well (please see Fig. 10). However, the error attained for OMC is little higher as compared to MDD in TR and TS phases, but the cramming of maximum number of observations is within the acceptable range as the wideness of section is obtained within ± 20% variations. This can also be confirmed from Fig. 12 that 80% and

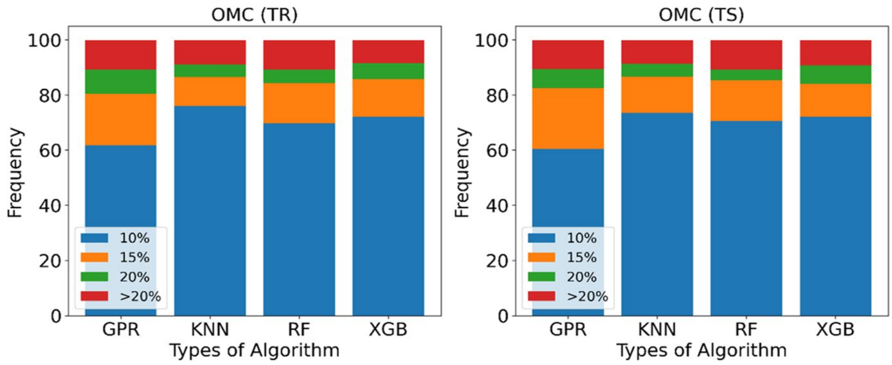


Fig. 12 Error frequency plot for GPR, KNN, RF, and XGB model in TR and TS dataset of OMC

90% observations for OMC can be predicted within ± 15% and ± 20% variations, respectively.

2.2 Regression Error Characteristic Curve

In the regression problems, regression error characteristic (REC) curves are equivalents to the receiver operating characteristic (ROC) curves as in classification problems. The *X*-axis and *Y*-axis of REC curve plot signify the error tolerance and accuracy, respectively, in terms of the percentage of points predicted within the tolerance (Kardani et al., 2021a; Asteris et al., 2021; Kardani et al., 2021b). An ideal model's curve should pass through the upper left corner and therefore should have an area under the curve (AUC) value that is 1. This means that the model can perfectly discriminate between all the positive and the negative class points. In general, an AUC of 0.5 suggests no discrimination, 0.7 to 0.8 is considered acceptable, 0.8 to 0.9 is deemed to be excellent, and more than 0.9 is considered outstanding. Figures 13 and 14 present the REC curve plot for the GPR, KNN, RF, and XGB model in the TR

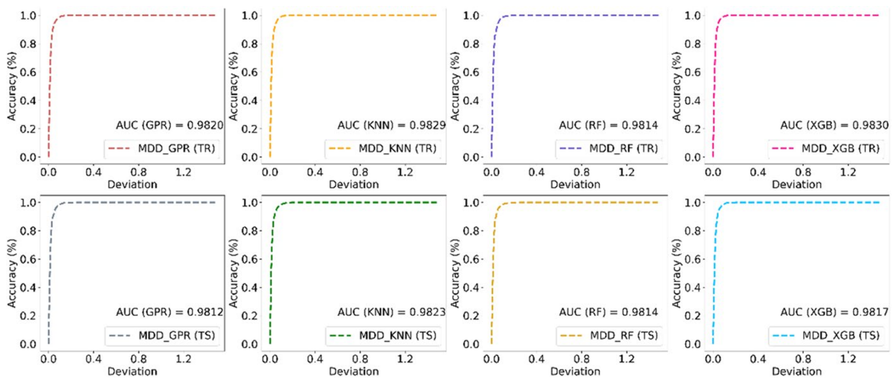


Fig. 13 REC plot for GPR, KNN, RF, and XGB model in TR and TS dataset of MDD

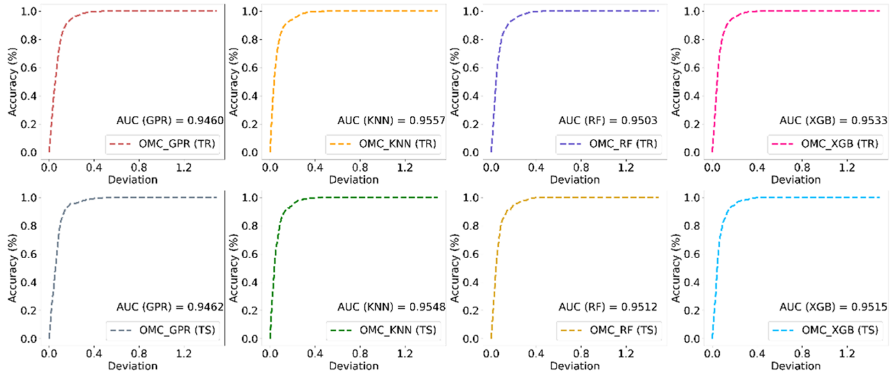


Fig. 14 REC plot for GPR, KNN, RF, and XGB model in TR and TS dataset of OMC

and TS dataset of MDD and OMC, respectively. As seen from Figs. 13 and 14, the REC curve in all cases exists in the upper left corner. Moreover, the AUC value obtained for all the models of MDD and OMC is greater than 0.9 which means that the developed models outperform very well and are specified to be reliable for predicting the MDD and OMC of soil.

2.3 Statistical Performance of Developed ML Models

Tables 6 and 7 present the statistical performance of MDD and OMC model, respectively, in the TR and TS phase for all four algorithms, i.e., GPR, KNN, RF, and XGB. As seen from Table 6, R^2 values obtained for GPR, KNN, RF, and XGB models in the TR phase are 0.681, 0.710, 0.635, and 0.749, respectively. This means that GPR, KNN, RF, and XGB models are sufficient to explain 68.1%, 71%, 63.5%, and 74.9% variability in MDD of TR dataset. Similarly, these models can explain

Table 6 Statistical performance of GPR, KNN, RF, and XGB model in TR and TS dataset of MDD

	GPR				KNN				RF				XGB			
	TR		TS		TR		TS		TR		TS		TR		TS	
R^2	0.681	2	0.618	1	0.710	3	0.669	3	0.635	1	0.634	2	0.749	4	0.674	4
R	0.825	2	0.788	1	0.843	3	0.820	3	0.798	1	0.798	2	0.873	4	0.824	4
MAE	0.043	2	0.046	1	0.040	3	0.042	4	0.046	1	0.046	2	0.040	4	0.044	3
RMSE	0.065	2	0.071	1	0.062	3	0.066	4	0.070	1	0.070	2	0.058	4	0.066	3
IOA	0.897	2	0.877	2	0.907	3	0.899	4	0.873	1	0.871	1	0.915	4	0.888	3
IOS	0.035	2	0.039	1	0.033	3	0.036	4	0.037	1	0.038	2	0.031	4	0.036	3
a5 index	0.891	2	0.890	4	0.898	3	0.886	3	0.877	1	0.885	1	0.906	4	0.885	2
Score	14		11		21		25		7		12		28		22	
Total score	25				46				19				50			
Rank	3				2				4				1			

Table 7 Statistical performance of GPR, KNN, RF, and XGB model in TR and TS dataset of OMC

	GPR				KNN				RF				XGB			
	TR		TS		TR		TS		TR		TS		TR		TS	
R^2	0.602	1	0.496	1	0.660	3	0.631	3	0.647	2	0.613	2	0.722	4	0.635	4
R	0.776	1	0.719	1	0.815	3	0.796	3	0.808	2	0.783	2	0.863	4	0.798	4
MAE	1.346	1	1.395	1	1.093	4	1.138	4	1.225	2	1.226	2	1.126	3	1.222	3
RMSE	2.162	1	2.470	1	2.001	3	2.112	3	2.038	2	2.163	2	1.808	4	2.102	4
IOA	0.860	1	0.833	1	0.881	3	0.883	4	0.872	2	0.864	2	0.902	4	0.874	3
IOS	0.175	1	0.196	1	0.164	3	0.169	4	0.165	2	0.173	2	0.149	4	0.171	3
a10 index	0.618	1	0.605	1	0.761	4	0.736	4	0.697	2	0.706	2	0.722	3	0.721	3
Score	7		7		23		25		14		14		26		24	
Total score	14				48				28				50			
Rank	4				2				3				1			

61.8%, 66.9%, 63.4%, and 67.4% variability in TS dataset. Similarly, for OMC, the GPR, KNN, RF, and XGB model can describe 60.2%, 66%, 64.7%, and 72.2% in TR and 49.6%, 63.1%, 63.3%, and 63.5% in TS dataset, respectively. The coefficient of correlation (R) value obtained for GPR, KNN, RF, and XGB models in TR and TS datasets of MDD is almost equal to or more than 0.80. Whereas, for OMC, it is better for XGB and KNN models. The R value 0.824 and 0.820 obtained for XGB and KNN models, respectively, in MDD and 0.798 and 0.796 in OMC reveals that the predicted value obtained through these models are substantially correlated with the actual values. The least value of MAE and RMSE value obtained for the developed model illustrates that MDD and OMC of soil can be predicted efficiently through gravel, fine content, and plasticity index of soil. According to Smith (Smith, 1986), Alavi and Gandomi (Alavi & Gandomi, 2011), Yildirim and Gunaydin (Yildirim & Gunaydin, 2011), Tenpe and Patel (Tenpe & Patel, 2020), Nayak et al. (Nayak et al., 2023), and Verma et al. (Verma et al., 2023), if R value is $\geq$ 0.80 and mean absolute error is least, then the developed model is thought to be efficient in predicting the respective parameters.

2.4 Ranking Analysis for Selecting the Best-Fitted ML Model

Numerous statistical performance indices were adopted to evaluate the performance of the models, and the conclusions can easily be drawn by comparing their values. However, the situation becomes more mysterious when the value of different performance indices describes their own best models. In that particular situation, the ranking analysis (RA) can be helpful to identify the best model as it provides the overall consideration of all the performance indices. According to RA, used by many researchers in the past (Kardani et al., 2021a; Asteris et al., 2021; Kardani et al., 2021b; Zhang et al., 2020), a maximum score of s (equal to the total number of corresponding models) is assigned to the model having the highest value in particular performance indices, minimum to the model with the lowest value and the score to the other intermediate models is assigned either in the ascending or descending order. Tables 6 and 7 tabulate the rank for each model as per their scoring value. It is observed that the XGB algorithm espoused maximum score in both situations, i.e., MDD and OMC, therefore assigned as first rank followed by other algorithms.

2.5 Robustness/Validation of Developed ML Models

Although the prediction model exhibits higher accuracy in the TR and TS phase, it cannot be considered reliable without assessing the generalization capability on some dataset that has not been used during the model development process. Therefore, a proper validation of model is always desirable for ensuring its robustness. It is always favorable to identify the accuracy of predictive model for the dataset obtained through different environments/conditions/setups. In the present study, two types of validation levels were espoused: first-level validation, using *K*-fold cross-validation approach, and second-level validation, the experimental dataset taken from the literature.

2.5.1 Level 1 Validation

The level 1 validation was performed through *K*-fold cross-validation (CV) approach. This study effectively utilized ten folds to evaluate the predictive capability of GPR, KNN, RF, and XGB models of MDD and OMC. Figures 15 and 16 demonstrate the validation results of MDD and OMC models, respectively. The average *R* value (mean of 10 folds) obtained for the respective GPR, KNN, RF, and XGB models indicates the robustness of model for the MDD and OMC dataset. Thus, it is perceived from Figs. 15 and 16 that the model developed through each ML algorithm is assumed to be robust.

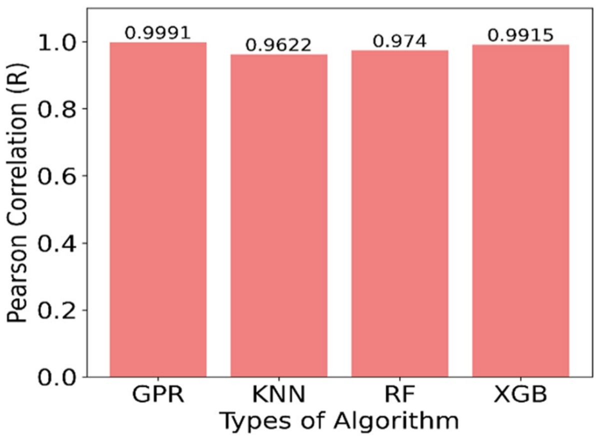


Fig. 15 Robustness of GPR, KNN, RF, and XGB models of MDD through K-fold CV

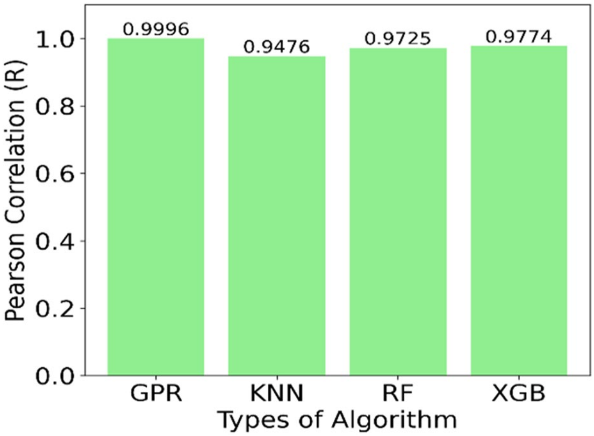


Fig. 16 Robustness of GPR, KNN, RF, and XGB models of OMC through K-fold CV

2.5.2 Level 2 Validation

For level 2 validation, the literature dataset which was obtained at similar laboratory testing conditions and different environmental and geological conditions was used for analysis. Therefore, this method is considered to be the most significant as well as reliable as it demonstrates the performance in terms of real-life implementation of the model on the field dataset which has not been seen by the model earlier. For this analysis, the only datasets were taken from Ramasubbarao and Sankar (Ramasubbarao & Sankar, 2013) and Bardhan et al. (Bardhan et al., 2021) study which exhibits the value of geotechnical parameters within the range of the present study dataset. Figure 17 shows the predicted performance of GPR, KNN, RF, and XGB models on the literature dataset in terms of error. As seen from Fig. 17, maximum observations of MDD and OMC can be predicted within ± 8% and ± 20% variations, respectively, which can also be confirmed from Fig. 18, showing the error frequency plot. Furthermore, it has also been perceived that MDD of soil can be predicted more reliably whereas OMC is within some acceptable ranges through the

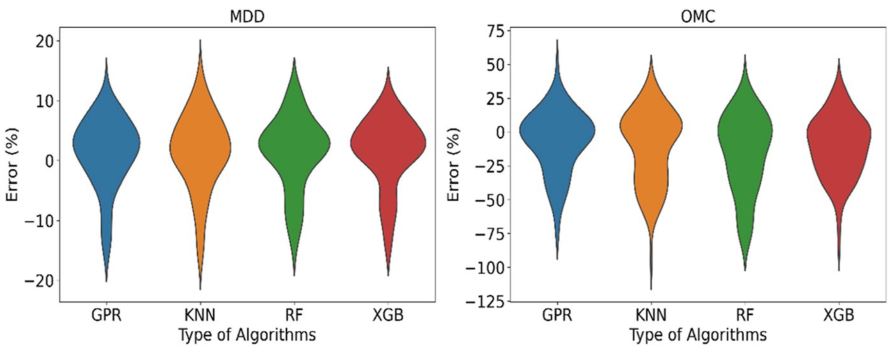


Fig. 17 Violin plot for GPR, KNN, RF, and XGB model for validation dataset

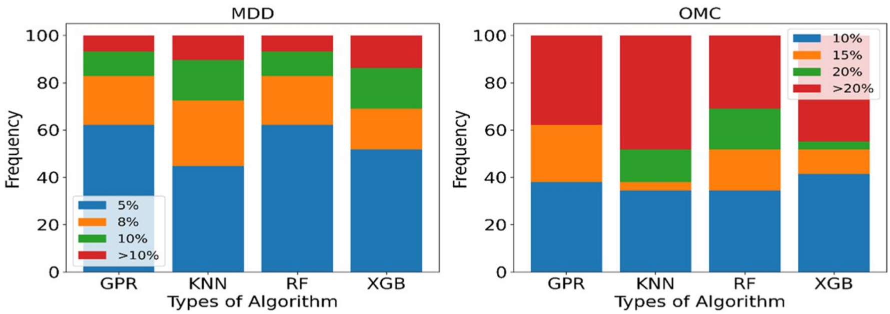


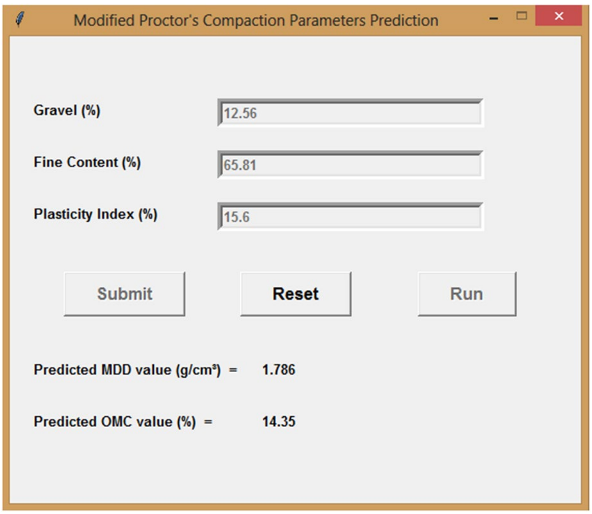
Fig. 18 Error frequency plot for GPR, KNN, RF, and XGB model for validation

adopted ML algorithms. However, further perfection in the model robustness can be accomplished by incorporating the dataset from other geological locations.

3 Development of MDD and OMC Prediction Software

In order to develop the prediction software, a graphical user interface (GUI) design technique was used which is a system of representing the features visually in the form of computer software. It is a simple and easy method to display the components according to user requirements. In this study, a reliable GUI was designed in python language for predicting the MDD and OMC of soils. The designed interface was designated as "*Modified Proctor's Compaction Parameters Prediction.*" Figure 19 depicts the structure of the designed interface. Initially, the values for input parameters, i.e., gravel (%), fine content, and plasticity index (%), are inserted. Ultimately, the predicted MDD and OMC values can be achieved directly after clicking

Fig. 19 Developed software for predicting the MDD and OMC of soils



the Run button. The developed interface is not only beneficial for the researchers but considerably user friendly for the site engineers.

4 Conclusion

This study employed 2148 in situ soil samples gathered from diverse global geological locations, to establish the prediction model for modified proctor compaction parameters of soil, essential for heavy earthwork construction projects. To achieve this, several machine learning algorithms were utilized, and their hyperparameters were fine-tuned using the PSO algorithm. The performance of developed hybrid model was assessed through statistical performance measurement indices such as R^2, R, MAE, RMSE, IOA, IOS, and a5 index. The order of accuracy of model, evaluated in terms of ranking analysis, was XGB > KNN > GPR > RF and XGB > KNN > RF > GPR for MDD and OMC, respectively. Notably, XGB and KNN models displayed strong significance, with R values exceeding 0.80 in both training (TR) and testing (TS) datasets for MDD and OMC. Furthermore, the violin plot and error frequency plot reveal that the MDD and OMC of soil can be predicted within ± 5% and ± 20% variations, respectively. Based on the trend plot and AUC value obtained in REC curve, the developed models were considered reliable and outperforming very well in predicting the MDD and OMC of soil. Apart from the good results obtained in TR and TS phases, the excellency and closeness of results achieved in level 1 and level 2 validation itself authenticate the worthy generalization capability of the established models. Lastly, the developed software exhibit suitability for aiding the site engineers in future MDD and OMC predictions.

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Data Availability The developed software can be downloaded from here https://drive.google.com/drive/folders/1E-sNvyF0CpiwDW34cFGMYkKLCMB4vhJb?usp=share_link (along with the guidelines and limitations). The dataset is available upon reasonable request to the corresponding author of the manuscript.

Declarations

Conflict of Interest The authors declare no competing interests.

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